RUSHIL BARVE

Computer Science & Computational Mathematics · University of Waterloo



About Me

Computer science student at the University of Waterloo with a record in informatics olympiads, Model UN and student leadership. Experienced in leading teams, running events, training peers and public speaking, with a growing focus on building technology that has real impact.

Contact

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LOCATION
Waterloo, Ontario

Skills

LANGUAGES
Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS
FRAMEWORKS
React, Vite, Tailwind CSS, FastAPI, Node.js, Express, Hono, Godot 4, Zustand
TOOLS
Git, GitHub Actions, SQLite, IndexedDB, Render, Redis, Reddit Devvit, Claude API, PayPal REST APIs

Waterloo Teams

- WARG (Waterloo Aerial Robotics): Autonomy and Embedded Flight Software teams
- UW Orbital: student satellite design team
- UW Game Dev · Computer Science Club · UW Data Science Club

EDUCATION

University of Waterloo — Expected 2031
Computer Science (Co-op) · Computational Mathematics

Delhi Public School – Modern Indian School, Doha — Graduated 2026
CBSE Senior Secondary, Computer Science

PROJECTS

CircuitGuard — Python · FastAPI · React · Three.js
Winner, Ampere – AI for Motor Safety · FormulaTech Hacks 2026
- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.

Whisper Chain — Godot 4 · GDScript
Puzzle game built on function composition
- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.
- ~3,000 lines of GDScript with UI-free game logic, ~1,500 test assertions including randomized property tests, and GitHub Actions CI building web and Windows exports.

Daydream — Godot 4 · GDScript · Python · NumPy
Puzzle game where progress happens only when you let go of the controls
- Puzzle game built on an inverted input mechanic: the dream world advances only while the player touches nothing; awake, they rearrange a desk whose side-on 3× projection is the dream level, so both worlds share one data model.
- Deterministic fixed-step (120 Hz) physics enables headless tests that replay a stored solution for all 8 levels and check each wins within par, run by a GitHub Actions workflow.

Lanternlight (in progress) — TypeScript · React · Tailwind · Hono · Redis · Devvit
Reddit Game of the Day hackathon: daily puzzle game on Devvit Web
- Daily mirror-and-lantern puzzle for Reddit: a seeded generator builds each day's board, and an exhaustive solver over every reachable mirror arrangement sets the true minimum taps (par).
- Server (Hono + Redis) replays each player's tap sequence against the canonical puzzle to verify solves, then updates a sorted-set leaderboard, score distribution, streaks and stats; a scheduler posts a new puzzle daily.

MUN Chair Toolkit — TypeScript · React · Vite · Tailwind · Zustand · Dexie
Local-first web app for Model UN chairs · live on GitHub Pages
- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.
- Kept rules in framework-free functions (src/lib/rules.ts) with automated tests; deployed to GitHub Pages.

PayPilot (in progress) — JavaScript · Node.js · Express · Claude API · PayPal REST APIs
PayPal AI Hackathon: AI back-office agent for freelancers
- Chat agent that turns plain-English requests into PayPal invoices, payment links, reminders and refunds through a Claude tool-use loop over 11 PayPal API tools.
- Human-in-the-loop safety: actions that email customers or move money pause the agent loop, save its state and resume after an Approve/Decline click.

Door Occupancy Counter (in progress) — Python · SQLite · JavaScript · TensorFlow.js
Prototype: live building occupancy from door sensors
- Counts people entering and leaving each door from two-beam sensor events (direction from beam order), keeping a live headcount with no manual logging.
- Dashboard compares today with the same weekday over the last 4 weeks and forecasts the rest of the day; adds a busiest-hours heatmap, capacity alerts and an emergency roll-call view.
- Tracks its own accuracy from hand counts and empty-building nights; restores the day's count from the SQLite event log after a restart; daily Excel reports written with no external libraries.

Continental Smash — Godot 4 · GDScript · GLSL shaders
UW Game Jam Fall 2026 entry (theme: Drift)
- Local two-player game where drifting continents ram each other to raise mountains and shove the rival into an ocean trench.
- Custom shaders for the planet, ocean, terrain and clouds, a narrated cinematic intro, particle impact effects and an Inspector-tunable ruleset.

Track Nector — Python · HTML · JSON · Render
Self-hosted personal finance tracker
- Lightweight tracker for card balances and daily spending built around manual entry, with no bank account linking.
- Chose self-hosting and manual input deliberately to keep financial data private.

Corso Website — HTML · Bootstrap · Sass · JavaScript
First web project, from a web design course
- Responsive site for courses and workshops: sticky navbar that changes on scroll, image carousel, registration and email-subscribe forms.
- Later cleaned up the course design myself: working forms with validation, accessibility fixes, ~70% smaller images; live at rsk0110.github.io/webproject.

Source code available **on request** · more projects at rsk0110.github.io/s2rkportfolio

HACKATHONS

FormulaTech Hacks — Winner (Ampere)	2026	RevenueCat Shipaton	2026
OpenCV AI Competition (AWS)	2026	Nebius x NVIDIA Global AI Hackathon	2026
IBM BOB 2.0 Hackathon	2026	DPS MIS Doha Hackathon	2025
Birla School Hackathon — Qatar	2024	CMU Qatar Hackathon	2023

8 hackathons · 5 won · 1 organised

Certifications

- Python Essentials 2 · HTML Essentials · Computer Hardware Basics
- Accenture Nordics – Software Engineering Job Simulation
- Bloomberg – Client Service Job Simulation

Honors & Awards

- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze Medal of Distinction (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Special Achievement Certificate, International Olympiad of English Language (2024-25)
- Best Delegate, UN Commission on Crime Prevention and Criminal Justice, Monarch MUN (2024-25)
- Honorary Mention, SDG 13, DPS-Modern Indian School MUN (2024-25)
- Six Best Delegate awards and one Honorable Mention across 8+ conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26); Bronze (2023-24, 2024-25); national-level athlete

Interests

- Basketball: house captain, national-level CBSE Clusters player
- Kickboxing (Feb 2026 – present)
- Swimming, chess, dance, visual art

LEADERSHIP & EXPERIENCE

Director of MUN & Electoral and Democratic Club — May 2025 – Jun 2026
DPS-Modern Indian School, Doha

- Led and managed the school's MUN program: organized conferences, trained delegates, promoted diplomacy and public speaking.

Head of Marketing, TEDxYouth@DPSMIS — Sep – Dec 2025

- Led branding, social media and promotion for a TEDx-licensed youth event, working across curation, design, operations and technical teams.

Head of Operations, DPSDQ MUN — Jun – Oct 2025

- Ran logistics and delegate services for the 25th-anniversary edition, coordinating with the Secretariat.

Team Leader, Qatar Children's Science Congress — Dec 2024 – Mar 2025

- Led an original student research project from experimental design to data analysis and expert-panel presentation.

Co-Founder, HuAr (Humanities.Arts) — Feb 2024 – Mar 2025

- Co-founded a student organization guiding students on humanities and arts paths; organized conferences and speaker sessions.

Head of Outreach & Logistics, Q-SOMA — Jun 2025 – Jun 2026
Qatar Society of Medical Aspirants

- Led outreach and logistics and supported event management for the society.

MUN Trainer & Head Chair — May 2024 – May 2026
DPS-MIS · Catalyst MUN · Global MUN Network

- Trained delegates in research, procedure and resolution drafting; Head Chair of Historical Security Council (Catalyst MUN) and WHO (Global MUN Network).

Event Manager, New Gen Network — Aug – Nov 2025

Event Coordinator, SFI Football Cup — May 2025

- Coordinated logistics and operations for a charity football cup.

Content Contributor, AetherMUN · Brand Ambassador, Luminos Foundation — May – Aug 2025

Education Mentor, DPS-Modern Indian School — May 2024 – Jan 2025

- Mentored a junior student while in Grade 11.

VOLUNTEERING

Student Volunteer, CMU-Qatar MindCraft Workshop — Nov 2024

- Assisted students with robotics and technology activities.

Student Volunteer, Embassy of India International Day of Yoga — Jun 2024

CERTIFICATES & RECOGNITION

MODEL UNITED NATIONS

UNHSC Head Chair — Catalyst MUN — 2025-26
Certificate of Appreciation, Director of MUN Club and Electoral & Democratic Club — DPS-Modern Indian School — 2025-26
Best Delegate — DPS-Monarch International School MUN — 2024-25
Honorary Mention — DPS-MUN — 2024-25
Participation as Delegate — DPS-MUN and DPS-Monarch MUN — 2024-25

OLYMPIADS & ACADEMICS

Gold Medal, International Informatics Olympiad — Silverzone Foundation — 2023-24
Special Achievement, International Informatics Olympiad — Silverzone Foundation — 2023-24, 2024-25
Special Achievement and Level 2, International Olympiad of English Language — Silverzone Foundation — 2024-25
Abacus Levels 1 and 2 (Excellence) and Abacus Challenge participation — TCA Qatar — 2018-19

SPORTS & CULTURAL

3rd Position, Basketball (Participation 2023-24) — CBSE Qatar Clusters — 2024-25
2nd Position, Basketball (3rd in 2022-23) — Sports Rendezvous, DPS-MIS — 2023-25
3rd Position, Group Dance — Sports Rendezvous, DPS-MIS — 2024-25
1st Position, Skipping Rope Challenge — Sports Rendezvous, DPS-MIS — 2021-22
Participation, Chess — Sports Rendezvous, DPS-MIS — 2022-23
Level 5 Swimming completion — Sports Castle Academy — 2023-24

OTHER EVENTS

Participation, Science India Forum (QCSC) — SIF Indian Cultural Centre, Embassy of India Qatar — 2024-25
Participation, Pariksha Pe Charcha — Ministry of Education, Government of India — 2024-25